



**Killara**  
HIGH SCHOOL

STUDENT NUMBER

|  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|

# HSC MATHEMATICS

## *COURSE*

ASSESSMENT TASK NO 3

Date: Friday 27/5/11

**TIME ALLOWED:**    *Reading time: 5 minutes*  
                              *Working time: 60 minutes*

**INSTRUCTIONS:**    This paper consists of 2 questions  
                              ANSWER all 2 questions  
                              Start each question in a new booklet  
                              Marks may not be given for untidy and poorly arranged work.  
                              All relevant working should be shown.

***THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM***

**OUTCOMES ASSESSED:**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1 - Appreciation of the scope, usefulness, beauty and elegance of mathematics<br>H1 - seeks to apply mathematical techniques to problems in a wide range of practical contexts<br>H2 - constructs arguments to prove and justify results<br>H3 - manipulates algebraic expressions involving logarithmic and exponential functions<br>H5 - applies appropriate techniques from the study of calculus & trigonometry to solve problems<br>H8 - uses techniques of integration to calculate areas and volumes<br>H9 - communicates using mathematical language, notation, diagrams and graphs |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                 | MARKS |
|-------------------------------------------------|-------|
| QUESTION 1 Trigonometric Functions              | /22   |
| QUESTION 2 Exponentials and logarithm Functions | /18   |
|                                                 | /40   |

*This assessment task constitutes 15% of the Higher School Certificate Course Assessment.*

QUESTION 1: (22 marks)

MARKS

a. Differentiate the following with respect to  $x$ :

i)  $\tan 3x$  1

ii)  $\cos^3 x$  1

iii)  $e^{2x} \sin x$  2

b. Find:

i)  $\int_0^{\pi} \sin \frac{x}{2} dx$  2

ii).  $\int_0^{\frac{\pi}{4}} (2 \sin x + \sin 2x) dx$  2

c. Consider the function  $y = 1 + \sqrt{3} \sin x + \cos x$ .

i) Find the equation of the tangent to the graph of the function 3

at  $x = \frac{5\pi}{6}$ .

ii) Find the maximum and minimum values of  $1 + \sqrt{3} \sin x + \cos x$   
in the interval  $0 \leq x \leq 2\pi$  3

d. Evaluate  $\int_0^{\frac{\pi}{3}} \sec^2 x dx$  by:

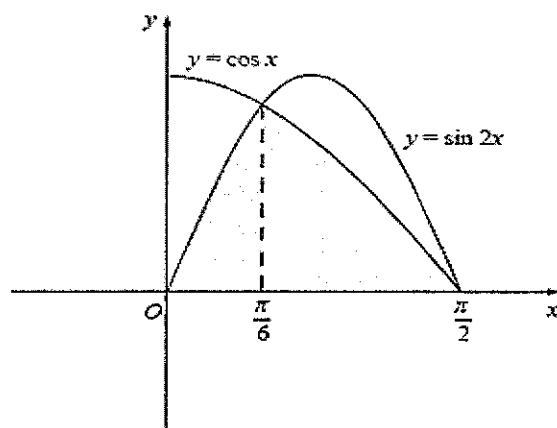
i) direct integration (give answer as exact value). 2

ii) using Simpson's rule using 3 function values( answer correct  
to 2 significant values). 2

e. The diagram shows the graphs of the functions  $y = \cos x$  and  $y = \sin 2x$  4

between  $x=0$  and  $x = \frac{\pi}{2}$ . The two graphs intersect at  $x = \frac{\pi}{6}$  and  $x = \frac{\pi}{2}$ .

Calculate the area of the shaded region.



QUESTION 2: (18 marks)

MARKS

a. Differentiate the following with respect to  $x$ :

i)  $e^{5x}$  1

ii)  $\frac{\ln x}{x}$  2

iii)  $\log_e(\cos 2x)$  2

b. Given that  $\log_a b = 2.75$  and  $\log_a c = 0.25$ , find the value of:

i)  $\log_a \left(\frac{b}{c}\right)$  1

ii)  $\log_a (bc)^3$  1

c. Evaluate  $\log_5 12$ , correct to 2 decimal places. 2

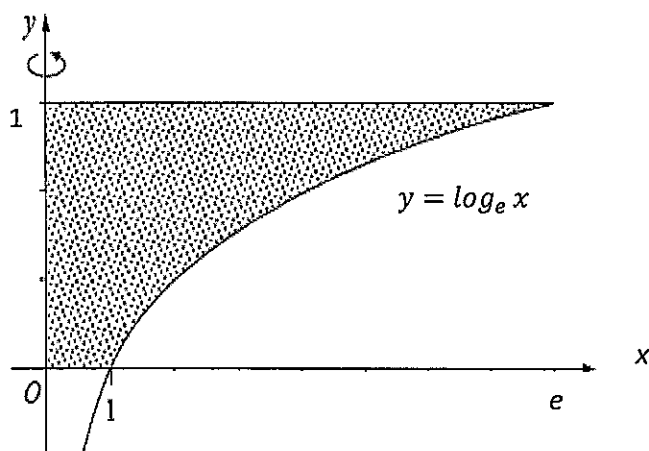
d. Find  $\int \frac{x-1}{x^2-2x} dx$ . 2

e. Evaluate i)  $\int_0^1 e^{4x} dx$ . 2

ii)  $\int_0^1 (e^{2x} + e^{-x}) dx$  2

f. The diagram shows the graph of  $y = \log_e x$  between  $x = 1$  and  $x = e$ . 3

The shaded region, bounded by  $y = \log_e x$ , the line  $y = 1$ , and the  $x$  and  $y$  axes, is rotated about the  $y$  axis to form a solid. Find the volume of this solid.



## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

# 2011 MATHEMATICS TASK 3 ANSWERS

## QUESTION 1

a i  $\frac{d}{dx} \tan 3x = 3 \sec^2 3x$

1 mark correct answer

ii  $\frac{d}{dx} \cos^3 x = -3 \sin x \cos^2 x$

1 mark correct answer

iii  $\frac{d}{dx} e^{2x} \cdot \sin x = 2e^{2x} \sin x + e^{2x} \cos x$   
 $= e^{2x} (2 \sin x + \cos x)$

u'v + v'u 1 mark for each correct part.

b i)  $\int_0^{\pi} \sin \frac{x}{2} dx = \left[ -2 \cos \frac{x}{2} \right]_0^{\pi}$   
 $= -2 \left( \cos \frac{\pi}{2} - \cos 0 \right)$   
 $= -2 (0 - 1)$   
 $= 2$

1 mark correct integration.

1 mark correct answer

ii)  $\int_0^{\pi/4} (2 \sin x + \sin 2x) dx = \left[ -2 \cos x - \frac{\cos 2x}{2} \right]_0^{\pi/4}$   
 $= - \left[ \left( 2 \cos \frac{\pi}{4} + \frac{1}{2} \cos \frac{\pi}{2} \right) - \left( 2 \cos 0 + \frac{\cos 0}{2} \right) \right]$   
 $= - \left[ 2 \frac{1}{\sqrt{2}} - 0 - \left( 2 + \frac{1}{2} \right) \right]$   
 $= -\sqrt{2} + \frac{5}{2}$

1 mark correct integration

1 mark correct substitution and answer

c i  $y = 1 + \sqrt{3} \sin x + \cos x$

$\frac{dy}{dx} = \sqrt{3} \cos x - \sin x$

1 mark differentiation and finding m or y

at  $x = \frac{5\pi}{6}$   $m = \sqrt{3} \cos \frac{5\pi}{6} - \sin \frac{5\pi}{6}$   
 $= \sqrt{3} \left( -\frac{\sqrt{3}}{2} \right) - \frac{1}{2}$   
 $= -2$

2 marks gradient, y and differentiation

at  $x = \frac{5\pi}{6}$   $y = 1 + \sqrt{3} \sin x + \cos x$   
 $= 1 + \sqrt{3} \sin \frac{5\pi}{6} + \cos \frac{5\pi}{6}$   
 $= 1 + \sqrt{3} \times \frac{1}{2} + \frac{\sqrt{3}}{2}$   
 $= 1$

$\therefore y - 1 = -2 \left( x - \frac{5\pi}{6} \right) \Rightarrow y = -2x + \frac{5\pi}{3} + 1$   
 eqn of tangent at  $\frac{5\pi}{6} = x$

3 marks equation of tangent

Question 1 continue

$$c ii \quad y' = \sqrt{3} \cos x - \sin x$$

$$y' = 0 \quad \sqrt{3} \cos x = \sin x$$

$$\tan x = \sqrt{3}, \quad x = \frac{\pi}{3}, \frac{4\pi}{3}, \text{ turning pts:}$$

$$x = \frac{\pi}{3} \quad y = 1 + \sqrt{3} \sin x + \cos x$$

$$= 1 + \frac{\sqrt{3}}{2} + \frac{1}{2}$$

$$= 3 \quad \therefore \left( \frac{\pi}{3}, 3 \right)$$

$$x = \frac{4\pi}{3} \quad y = 1 + \sqrt{3} \sin \frac{4\pi}{3} + \cos \frac{4\pi}{3}$$

$$= 1 + \sqrt{3} \left( -\frac{\sqrt{3}}{2} \right) + \left( -\frac{1}{2} \right)$$

$$= -1 \quad \therefore \left( \frac{4\pi}{3}, -1 \right)$$

Nature of turning points

$$y'' = -\sqrt{3} \sin x - \cos x$$

$$\text{at } x = \frac{\pi}{3}$$

$$y'' = -\sqrt{3} \sin \frac{\pi}{3} - \cos \frac{\pi}{3}$$

$$= -\sqrt{3} \cdot \frac{\sqrt{3}}{2} - \frac{1}{2}$$

$$= -2 < 0 \text{ concave down, max point.}$$

$$\text{at } x = \frac{4\pi}{3} \quad y'' = -\sqrt{3} \sin \frac{4\pi}{3} - \cos \frac{4\pi}{3}$$

$$y'' = -\sqrt{3} \left( -\frac{\sqrt{3}}{2} \right) - \frac{-1}{2}$$

$$= 2 > 0 \text{ conc. up. min pt.}$$

$$\therefore \left( \frac{\pi}{3}, 3 \right) \text{ max point, } \left( \frac{4\pi}{3}, -1 \right) \text{ min. point.}$$

$$d i \quad \int_0^{\pi/3} \sec^2 x dx = \left[ \tan x \right]_0^{\pi/3}$$

$$= \tan \frac{\pi}{3} - \tan 0$$

$$= \sqrt{3}$$

$$ii \quad h = \frac{\pi/3 - 0}{2} = \frac{\pi}{6} \quad \sec^2 x = \frac{1}{\cos^2 x}$$

| $x$        | 0 | $\pi/6$       | $\pi/3$ |
|------------|---|---------------|---------|
| $\sec^2 x$ | 1 | $\frac{4}{3}$ | 4       |

$$1 \text{ mark } \tan x = \sqrt{3}$$

1 mark turning points

1 mark correctly finding nature of turning points

1 mark correct integration

1 mark for  $\sqrt{3}$ 1 mark finding  $h$  &

$$\frac{h}{3} = \frac{\pi}{18}$$

du continue

$$\int_0^{\pi/3} \sec^2 x dx = \frac{\pi/6}{3} \left[ \sec^2 0 + 4 \sec^2 \frac{\pi}{6} + \sec^2 \frac{\pi}{3} \right]$$

$$= \frac{\pi}{18} \left[ 1 + 4 \times \frac{4}{3} + 4 \right]$$

$$= 1.8$$

1 mark correct substitution  
and

~~1 mark~~ correct answer

$$e) A_2 = \int_0^{\pi/6} \sin 2x dx + \int_{\pi/6}^{\pi/2} \cos 2x dx$$

$$= \left[ -\frac{\cos 2x}{2} \right]_0^{\pi/6} + \left[ \sin x \right]_{\pi/6}^{\pi/2}$$

$$= -\frac{1}{2} \left[ \cos \frac{\pi}{3} - \cos 0 \right] + \left[ \sin \frac{\pi}{2} - \sin \frac{\pi}{6} \right]$$

$$= -\frac{1}{2} \left[ \frac{1}{2} - 1 \right] + \left[ 1 - \frac{1}{2} \right] \Rightarrow -\frac{1}{2} \times -\frac{1}{2} + \frac{1}{2}$$

$$= \frac{3}{4} \text{ unit}^2$$

2 mark correct boundaries  
and functions

1 mark correct integration

1 mark correct answer  
working out  
and

## QUESTION 2

a) i)  $\frac{d}{dx} e^{5x} = 5e^{5x}$

ii)  $\frac{d}{dx} \frac{\ln x}{x} = \frac{\frac{1}{x} \cdot x - \ln x}{x^2}$

$$= \frac{1 - \ln x}{x^2}$$

iii)  $\frac{d}{dx} \log_e \cos 2x = \frac{-\sin 2x \cdot 2}{\cos 2x}$

$$= -2 \tan 2x$$

b) i)  $\log_a \frac{b}{c} = \log_a b - \log_a c$

$$= 2.75 - 0.25$$

$$= 2.50$$

ii)  $\log_a (bc)^3 = 3 \log_a bc$

$$= 3 [\log_a b + \log_a c]$$

$$= 3 [2.75 + 0.25]$$

$$= 9$$

c)  $\log_5 12 = \frac{\log_e 12}{\log_e 5}$

$$= 1.54$$

(3)

Question 2 continue

$$d) \int \frac{x-1}{x^2-2x} dx$$

$$f(x) = x^2 - 2x \quad f'(x) = 2x - 2$$

$$\int \frac{2(x-1)}{2(x^2-2x)} dx = \frac{1}{2} \int \frac{2x-2}{x^2-2x} dx$$

$$= \frac{1}{2} \ln(x^2-2x) + C$$

(you can ignore C)

$$e) i) \int_0^1 e^{4x} dx = \left[ \frac{e^{4x}}{4} \right]_0^1$$

$$= \frac{1}{4} [e^4 - 1]$$

$$ii) \int_0^1 (e^x + e^{-x}) dx =$$

$$= \left[ e^x \frac{1}{2} + -e^{-x} \right]_0^1$$

$$= \left[ \frac{e^2}{2} - e^{-1} \right] - \left[ \frac{e^0}{2} - e^0 \right]$$

$$= \frac{e^2}{2} - \frac{1}{e} - \frac{1}{2} + 1$$

$$= \frac{e^2 + e - 2}{2e}$$

$$f) y = \log_e x$$

$$e^y = e^{\log_e x}$$

$$x = e^y$$

$$V = \pi \int_0^1 x^2 dy$$

$$= \pi \int_0^1 e^{2y} dy$$

$$= \pi \left[ \frac{e^{2y}}{2} \right]_0^1$$

$$= \frac{\pi}{2} [e^2 - e^0]$$

$$= \frac{\pi}{2} [e^2 - 1] \text{ unit}^3$$